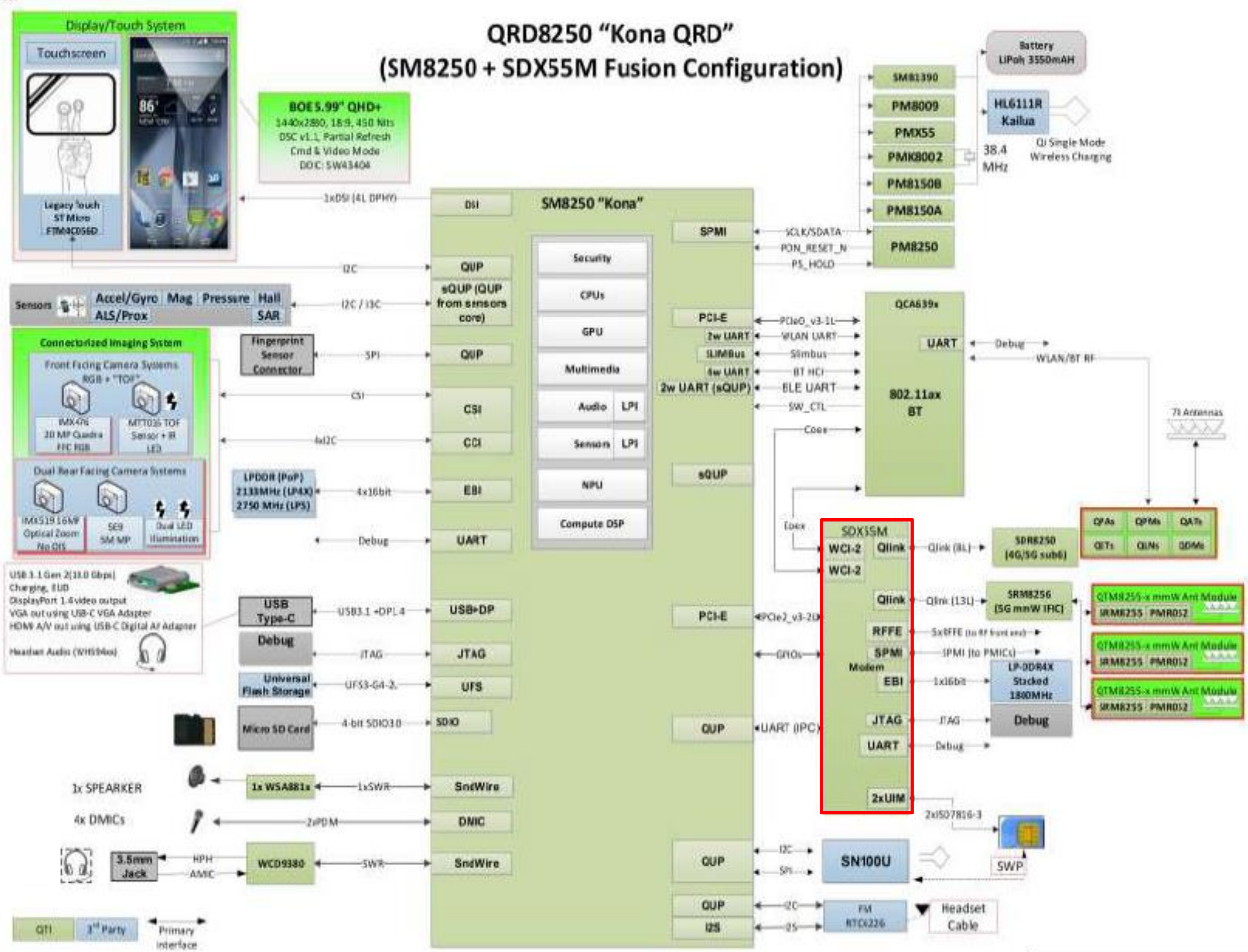
ZS670KS Tequila
Baseband Trouble shooting

Agenda

- System Block Diagram
- Phone Introduction
- PCB Introduction
  - -Main Board (MB)
  - -SIM Sub-Board (SIM SB)
  - -USB Sub-Board (USB SB)
  - -Camera Sub-Board (CAM SB)
- Repair Guideline

System Block diagram



Phone Introduction
Back side (1/2) – MB & USB SB

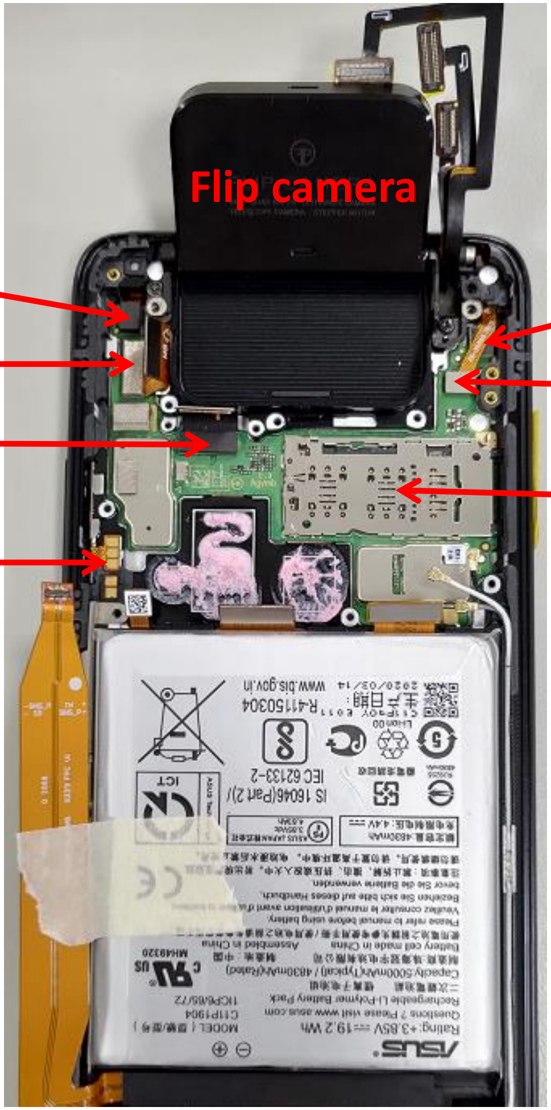
Phone Introduction
Back side (2/2) – SIM SB

Right side P/L sensor con. (J202)

SIM FPC2 con. (J101)

SIM FPC1 con. (J106)

Side key contacted pads

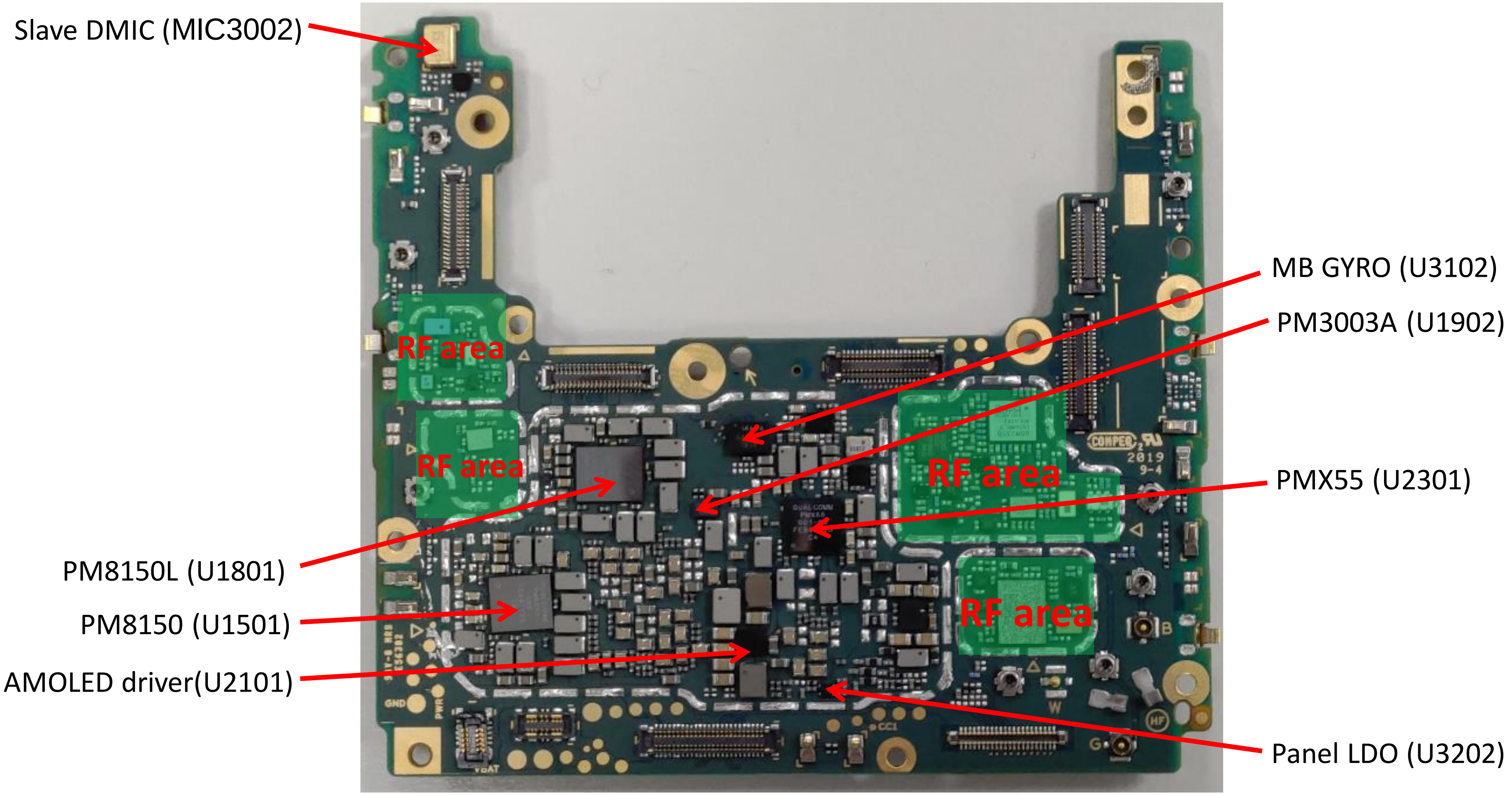


Left side P sensor con. (J203)

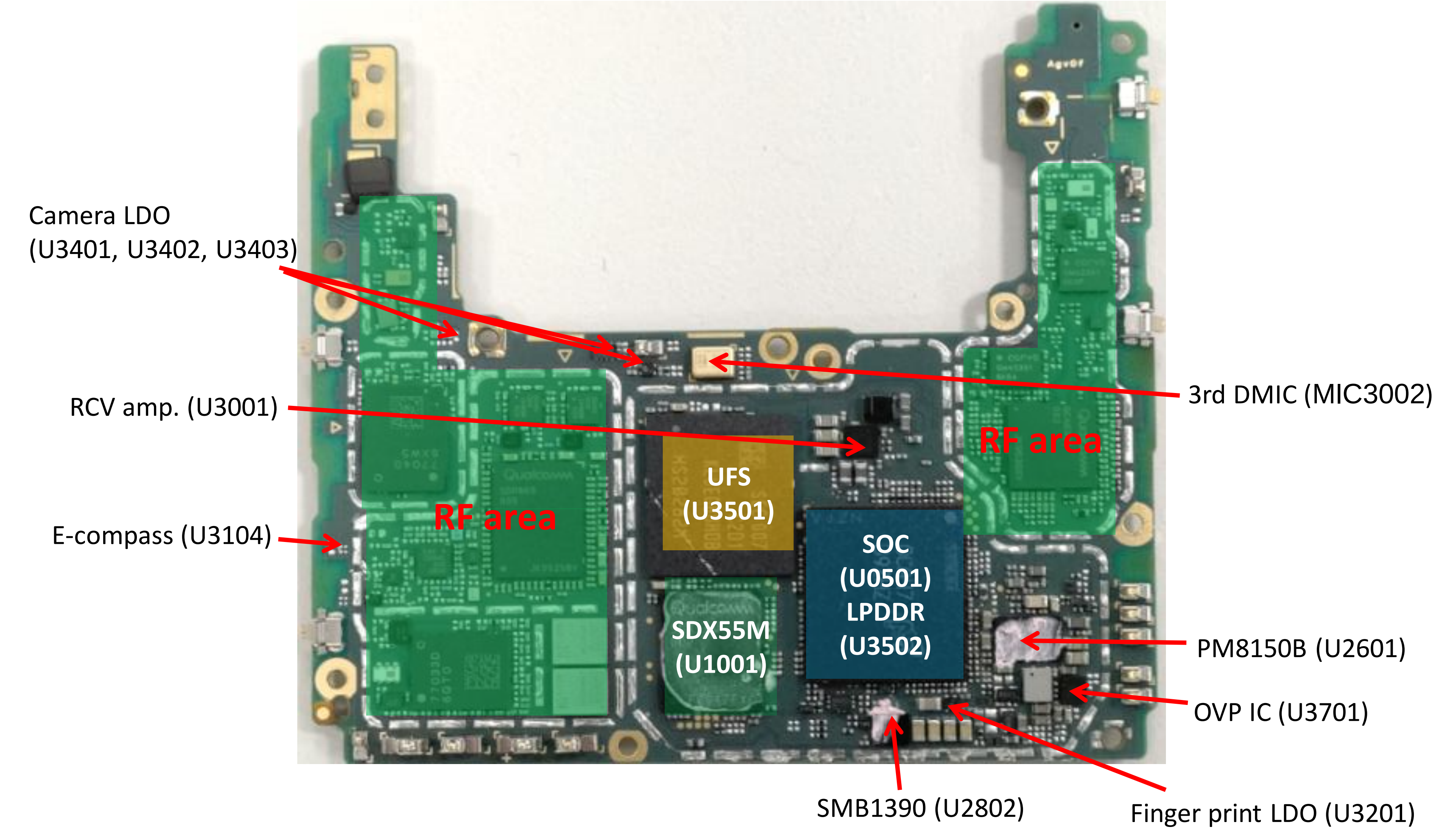
Angle sensor FPC con. (J201)

SIM/SD card tray con.

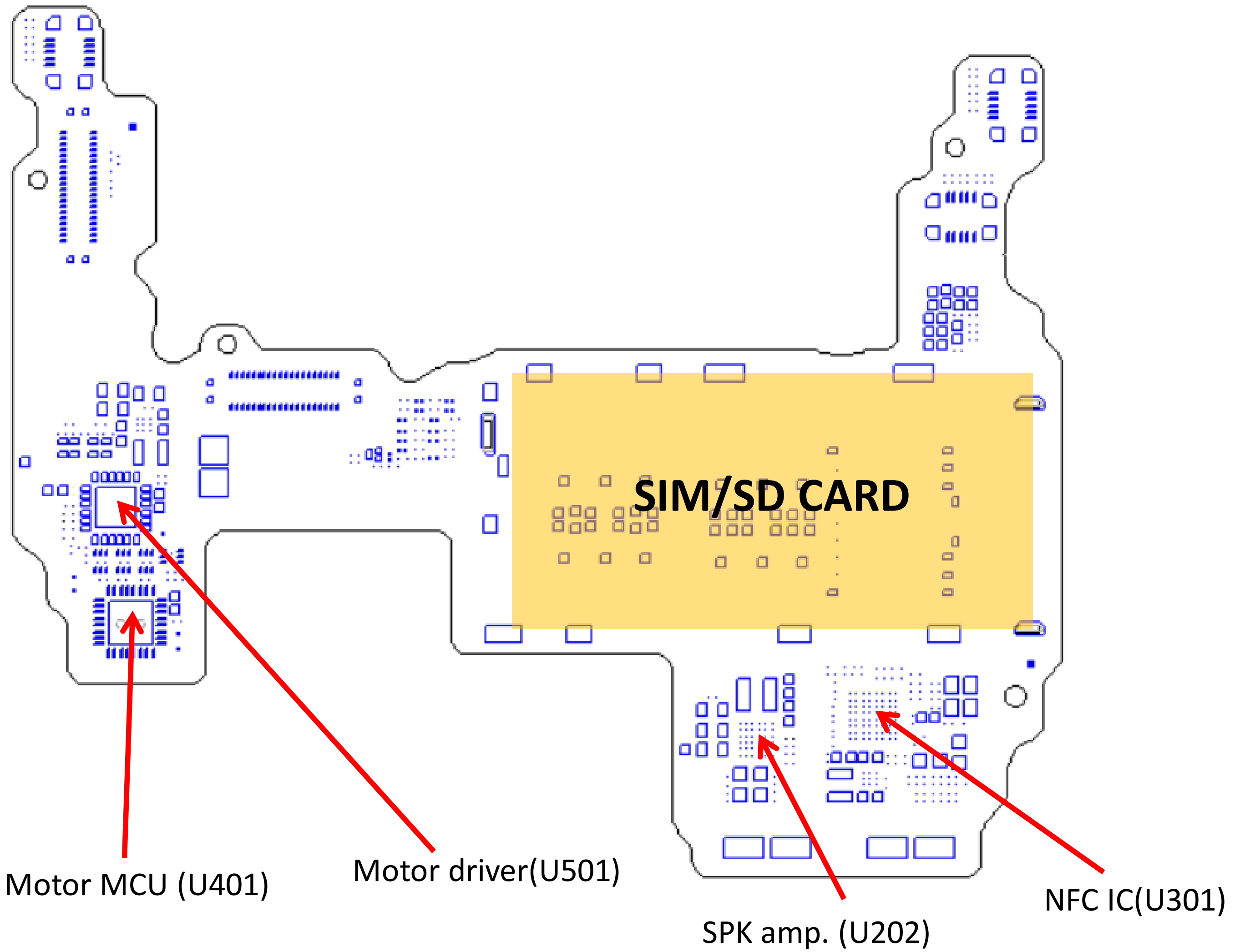
PCB Introduction
MB (1/2) - Bottom



PCB Introduction
MB (2/2) - Top



PCB Introduction
SIM SB Top (1/1)

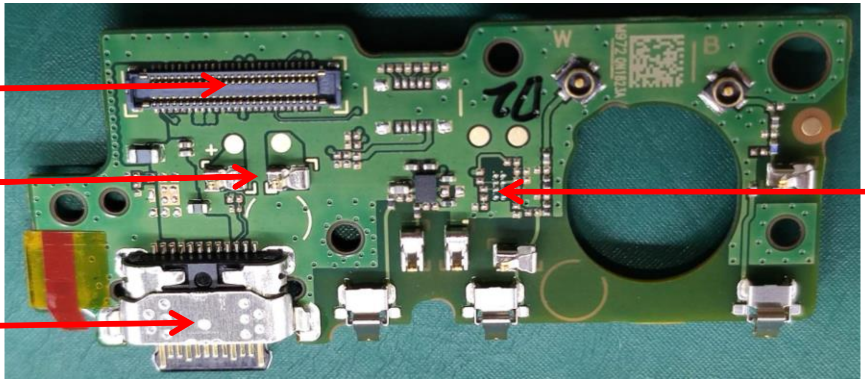


PCB Introduction
USB SB Top / Bottom (1/1)

IO Link FPC con. (J102)

SPK spring (ANT201, ANT202)

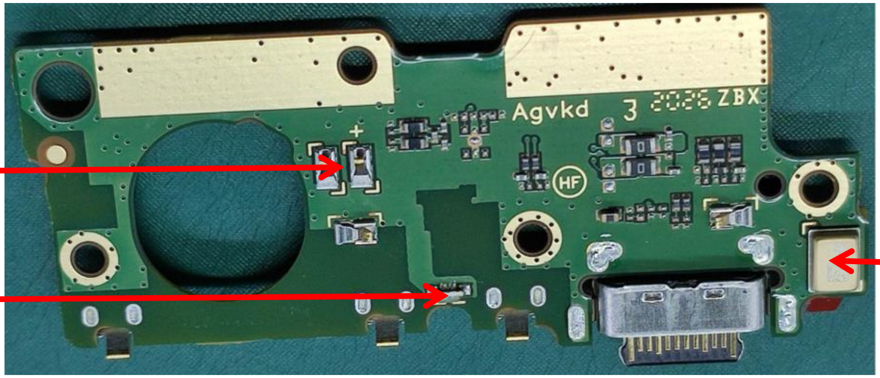
USB type C con. (J101)



SAR sensor (SAR401)

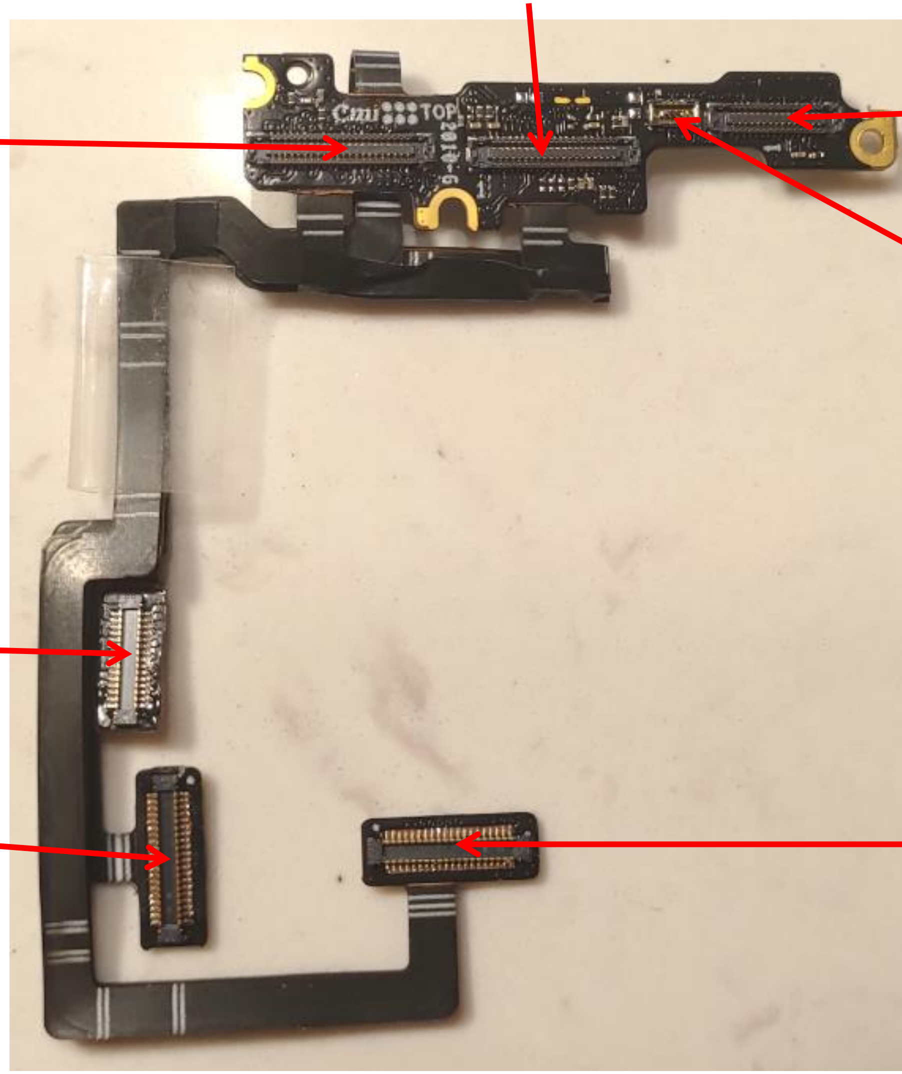
Vibrator spring (ANT101, ANT102)

Indicator LED (LED201)

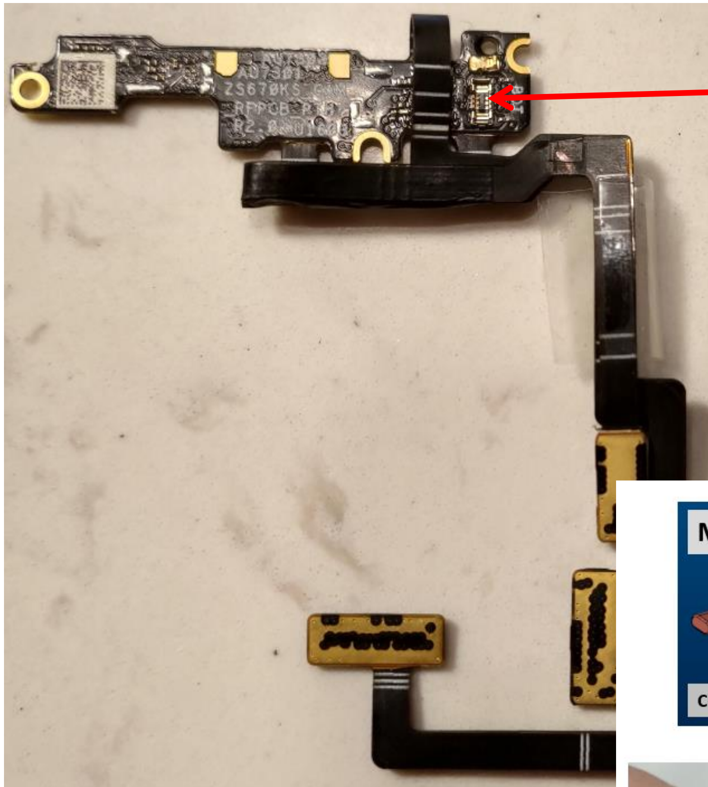


Main DMIC (MIC201)

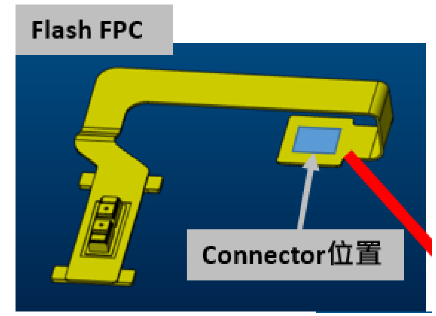
PCB Introduction
Camera SB Top (1/2)

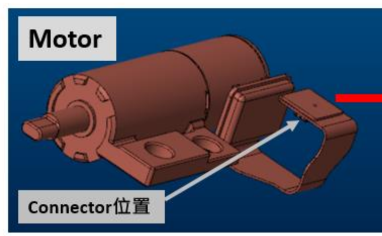


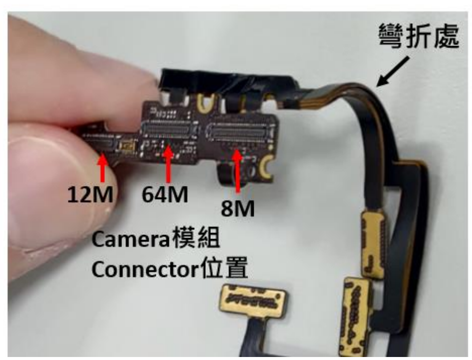
PCB Introduction
Camera SB Bottom (2/2)



Flash LED con. (CON1605)







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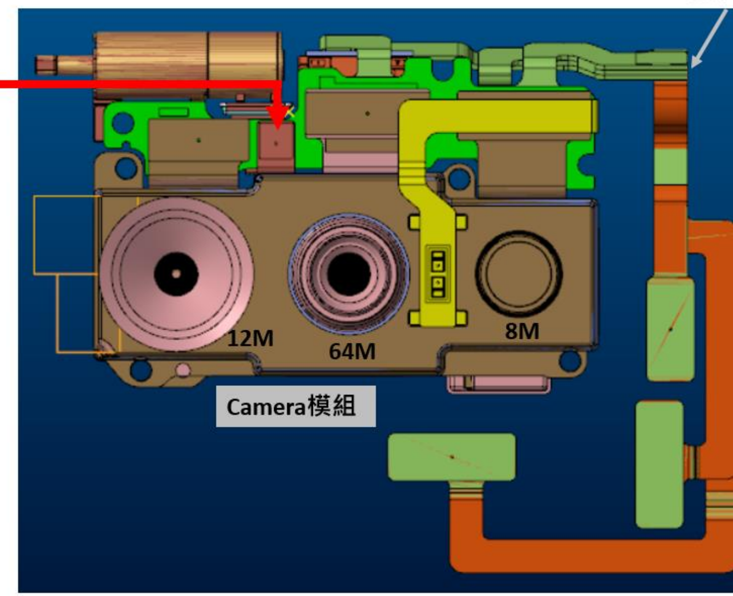


Repair Guideline

- Power on Fail
- UFS Fail
- Panel Fail
- Touch Fail
- 1st G+G Sensor Fail
- 2nd G+G Sensor Fail
- E-compass Sensor Fail
- Proximity Sensor Fail
- Finger Print Fail
- 64M Main Camera Fail
- 12M Wide Camera Fail
- 8M Tele Camera Fail
- SIM/SD Card Fail
- Vibrator Fail
- VOL Key Fail
- Motor Fail
- Indicator LED Fail
- Flash LED Fail
- Receiver Fail
- Speaker Fail
- Main Mic Fail
- Slave Mic Fail
- 3rd Mic Fail
- Headset Fail

Power ON Fail(1/3)

1. Check whether the battery voltage is correctly sent to the mainboard

-> Attach the battery to check if the TP3838 has voltage

1-1 No voltage: Check whether the battery is abnormal, or find the defect with the mainboard (whether there is a problem with the pin on the Ex. battery con (J3602))

1-2 There is voltage: check whether the battery voltage is above 3.4V, if it is a normal boot voltage, check whether the PS_HOLD (TP202) is pulled to maintain 1.8V when the power key is pressed or the USB is plugged in.

1-2-1. If PS_HOLD is pulled 1.8, check whether it is a screen problem or UFS abnormality

1-2-2.PS_HOLD can't hold to confirm the following:

2.VPH_PWR voltage -> Check J3401.43/J3401.44

1. Short-circuit to ground.According to x-ray, confirm whether the following components are connected with tin: J3401, U1801, U1501, U1902, U2101, U2301, U2201, U1401, U2601, U2802, U4501 and

   Whether the parallel components and RLC on the VPH_PWR path are abnormal, U502, J106, U301, U302 on SIM SB

2. No short circuit to ground

   Connect Power ON Fail (2/3)

Power ON Fail(2/3)

VPH_PWR is close to the battery voltage, MSM_PS_HOLD has no voltage -> 1. Confirm whether the side KEY works normally

(1) Confirm whether the side KEY FPC is good. (2) While pressing the Power key, confirm whether the voltage of TP3833 is 0V.

If not, check whether the SMT status of ANT3603 or the overlap with key shrapnel is normal

2. If the side KEY works normally, check Power on sequence

(1) Starting from VREG_BOB, measure the voltage of each power rail in sequence to determine whether the voltage is up. If any power rail is checked to have a short to ground, confirm the SMT status of all components on the path

(2) If there is no short-circuit to ground, but one or more groups have no voltage or abnormal voltage, check the SMT status of all components on the power path

(3) If the SMT status of all the originals on the power path of the group is completely normal, reheat or replace the PMIC in order (i) reheat U1501

(ii) reheat U1902

(iii) reheat U1801

(iv) reheat U2301

(v) Replace U1501

(vi) Replace U1902

(vii) Replace U1801

(viii) Replace U2301

After the hardware boot process is completed, MSM_PS_HOLD(T1911) will be 1.8V. If there is still no display, there are two possibilities:

UFS Fail→Replace CPU/DDR/UFS

OLED Fail-->Check whether the panel con (J3201) and panel are abnormal

Power ON Fail(3/3)

Power on sequence

Table 3-18 PM8250 power-on sequence

Event [1]	Device name	Timeline for KPD PON (ms) [2]
PON trigger received		0.00
BOB	PM8150L	50.23
S5A	PM8250	50.73
S6A	PM8250	50.99
S3C	PM8150L	51.30
L4A	PM8250	51.58
S8C	PM8150L	51.97
S6C	PM8150L	52.30
S3A_S2A_S1A	PM8250	52.55
L11A	PM8250	52.79
L9A	PM8250	53.12
L5A	PM8250	53.38
L18A	PM8250	53.64
S4A	PM8250	54.23
SLEEP_CLK	PM8250	54.30
L12A	PM8250	54.43
VREF_MSM [3]	PM8250	54.45
LN_BB_CLK1	PM8250	55.16
S1H	PM3003A	61.32
L3A	PM8250	62.01
S7C	PM8150L	62.24
L2A	PM8250	62.53
L17A	PM8250	62.99
L6A	PM8250	63.32
L6C	PM8150L	63.59
L9C	PM8150L	63.90
S10A	PM8250	64.37
S4E	PMX55	102.50

Table 3-18 PM8250 power-on sequence (cont.)

Event [1]	Device name	Timeline for KPD PON (ms) [2]
S3E	PMX55	102.75
S2E	PMX55	103.08
S7E	PMX55	103.32
S5E	PMX55	103.55
L6E	PMX55	104.32
L15E	PMX55	104.70
SLEEP_CLK_E	PMX55	104.81
L1E	PMX55	105.33
L5E	PMX55	108.83
L2E	PMX55	109.08
L14E	PMX55	109.31
L4E	PMX55	109.56
L10E	PMX55	109.92
PON_RESET_N_E	PMX55	109.98
PON_RESET_N [4]	PM8250	127.31

1. The power-off sequence is the inverse order of the power-on sequence.
2. The timing information is typical and should be used only for general reference
3. VREF_MSM does not have any internal pull downs and may discharge slowly depending on load current
4. PM8250 will abort the power-on sequence and shutdown if PS_HOLD is not asserted **high** by the processor within 200 ms of PON_RESET_N being asserted **high**

UFS Fail

1. Check whether the voltage supplied to UFS is normal

Power source	NET name	Check point	Voltage
VCC	VREG_L17A_2P96	C3501.1	2.95
VCCQ1	VREG_L6A_1P2	C3504.1	1.2

2. If the voltage is normal, reheat or replace the IC in sequence
   (i) Reheat U3501 (UFS)
   (ii) Reheat U201 (CPU/DDR)
   (iii) Swap U3501
   (iv) Swap U201

Panel Fail

After the Panel cross test, if the Panel body is confirmed to be normal, first check the SMT status of J3201.

Observe whether the panel unit's own cable and CON are damaged or defective

Then check the power supply status.

■ VREG_L11C_3P0_CON: Check J3201.2, C2118, R2109, C2002, U1801, C2117 to confirm whether it is abnormal.

■ VREG_L14A_1P8_CON: Check J3201, C2123, R2111, C2122, C2121, U3202, C3231, U1501, and confirm whether it is abnormal.

■ PANEL_1P2: Check C3230.1, U3202.1, J3201.9 to confirm whether it is abnormal.

■ VREG_DIS_-2P7_CON: Check J3201.3, C2116, R2107, U2101, C2107, C2108, C2115 to confirm whether it is abnormal

■ VREG_DIS_4P6_CON: Check J3201.43/41/39, C2114, R2108, C2105, C2106, U2101, C2113 to confirm whether it is abnormal

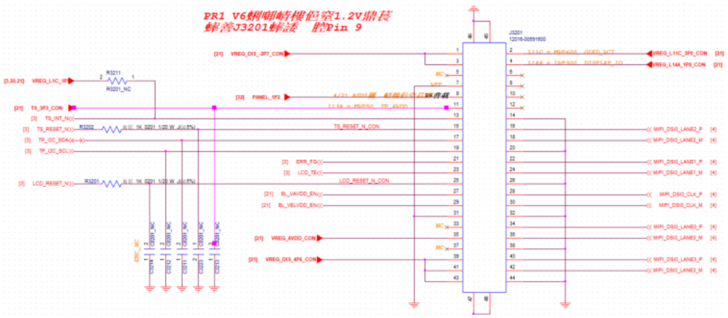
■ VREG_AVDD_CON: Check J3201.35, C2120, R2110, C2119, U2101, C2104 to confirm whether it is abnormal

If the above components are confirmed to be no abnormality, but one group\multiple groups of voltages are still abnormal, please Reheat U2101, U1501, U1801, U3201 in order. If not resolved, replace U2101, U1501, U1801, U3201 in order

Touch Fail

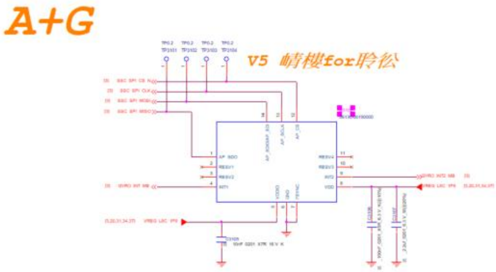
First confirm that the panel is a good product

1. First confirm the SMT status of J3201.
2. First check the part to be confirmed on the LCM failure page
3. Confirm the TP_I2C_SDA status and check the R305 SMT status.
4. Confirm the TP_I2C_SCL status and check the R301 SMT status.
5. Confirm TS_RESET_N status, check R3202, U201 X-ray to confirm whether it is abnormal.
6. Confirm the status of TS_INT_N. If there is an abnormality, it is a U201 problem.
7. Confirm TS_3P3_CON status, check R2113, C2125, C3213, J3201.11, C2126, U1501
8. If no problems are found in the above, please reheat U1501&U201 in order first, if it is invalid, replace U1501&U201 in order.



1st G+G Sensor Fail

1. Confirm whether VREG_L8C_1P8 has power supply, and measure C3105.2.
2. If there is a short circuit to ground or no voltage, first confirm the state of the capacitor and resistor connected to VREG_L8C_1P8.
3. If none of the above components are abnormal, but VREG_L8C_1P8 has no voltage, if it is short-circuited, find the short-circuit point and repair it, if it is not short-circuited, reheat U1801, if it is invalid, replace U1801
4. If VREG_L8C_1P8 has voltage, take x-ray to U3102 to confirm whether it is correctly mounted.
5. Measure whether the SSC_SPI_CLK, SSC_SPI_MOSI, SSC_SPI_MISO, SSC_SPI_CS_N and GYRO_INT1/2 signals are normal.
6. After finishing the above tests, it can't work, just replace U3102.
7. The above 1~6 points are the MB G-sensor confirmation process, and the G-sensor on the camera SB shall be handled by comparison



2nd G+G Sensor Fail

1. First confirm whether MB J3301, J3302, and J3303 all correspond to camera SB CON1601, CON1602, CON1603 buckle normally, and determine whether the problem occurs in CAM SB or MB
2. ■ If the problem occurs in CAM SB
3. Check the SMT status of CAM SB CON1703.
4. Measure whether R1724 is 2.9V, if not, check whether C1718.1 is shorted to ground.
5. Measure whether R1722 is 1.8V, if not, check whether C1716.1, C1712.1, C1707.1 are shorted to ground.
6. If none of the above is abnormal, please replace the 8M tele module, if it is invalid, replace the CAM SB.

7. ■ If the problem occurs in MB
8. Check the SMT status of MB J3301, J3302, J3303, C3338, C3301, C3330, R3301, C3318, C2005.
9. If none of the above is abnormal, first reheat MB U1801, U201, if it is invalid, replace U1801, U201.

E-compass Sensor Fail

1. Confirm whether VREG_L8C_1P8 has power supply, and measure C3108.1.
2. If there is a short circuit to ground or no voltage, first confirm the state of the capacitor connected to VREG_L8C_1P8.
3. If none of the above components are abnormal, but VREG_L8C_1P8 has no voltage, if it is short-circuited, find the short-circuit point and repair it, if it is not short-circuited, reheat U1801, if it is invalid, replace U1801.
4. If VREG_L8C_1P8 has voltage, take x-ray to U3106 to confirm whether it is correctly mounted.
5. Measure whether the SSC_I2C1_SCL, SSC_I2C1_SDA signals are normal.
6. After completing the above tests, you can't work, so replace U3106 directly.

Proximity Sensor Fail

1. Confirm whether SIM FPC1 J106, SIM FPC2 J101 are properly buckled with the mainboard, and check the SMT status.
2. Make sure that the two P sensors of SIM SB are connected to J202 and J203 normally.
3. Confirm which P sensor is abnormal J202 or J203

■ SIM SB J202:

1. View the SMT status of J202, R211, C212, C214, C216, C218, C228.
2. If there is no abnormality, measure whether there is 3.0V voltage at both ends of R211. If there is no abnormality, check whether the mainboard C2003.1 has 3.0V voltage. If there is no abnormality, reheat U1801 and replace U1801 invalid.
3. If the voltage is not abnormal, then confirm whether the AL/P_I2C_SDA_1/AL/P_I2C_SCL_1 path signal is abnormal.
4. If there is no abnormality, replace the P sensor connected to J202. If it is invalid, reheat U201 and replace U201 again.

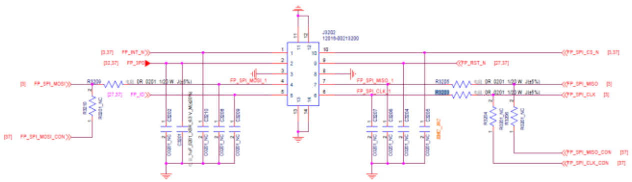
■ SIM SB J203:

1. Check the SMT status of J203, R212, C213, C215, C217, C219, C226.
2. If there is no abnormality, measure whether there is 3.0V voltage at both ends of R212. If there is no abnormality, check if there is 3.0V voltage on the mainboard C2003.1. If there is no abnormality, reheat U1801 and replace U1801 invalid.
3. If the voltage is not abnormal, then confirm whether the AL/P_I2C_SDA_2/ AL/P_I2C_SCL_2 path signal is abnormal.
4. If there is no abnormality, replace the P sensor connected to J202. If it is invalid, reheat U201 and replace U201 again.

Finger Print Fail

1. Confirm that the connection between MB J3202 and FP unit is normal.
2. Confirm whether C3202.1 has 3.0V voltage.
3. If there is a short circuit to ground or no voltage, first confirm the SMT status of U3201, C3229, C3228, C3201, R3205, R3203, C3207, C3206, C3204, C3205.
4. If none of the above components are abnormal and FP_3P0 has no voltage, confirm whether VREG_BOB has electricity, if it is short-circuited, find the short-circuit point and repair it, if it is not short-circuited, first reheat U1801, or if it is invalid, change U1801
5. If FP_3P0 has voltage, check J3202 whether the pin is abnormal.
6. Measure whether FPS_SPI_MOSI & FPS_SPI_MISO & FPS_SPI_CLK & FPS_SPI_CS_N & FPS_RST_N & FPS_INT_N & FP_ID signals are normal.
7. After the above tests are completed, the work cannot be done. Priority is given to replacing the FP with good products, and then reheating U201 if it is invalid, or replacing U201 if it is still invalid.

FPC



Angel sensor fail

1. Check whether the SIM board connector J201 has snapping problems or damage.
2. Check whether the Angel sensor FPC is broken or abnormal, and whether the IC is damaged.
3. Swap Angel sensor FPC/SIM board/SIM_Left FPC/MB confirms the phenomenon follower.

If it is connected to the SIM board or mainboard ：

1. Confirm that R208.2 has a 3.3V voltage. If not, confirm the SIM board R410, R406, R407, R405, R404, R402, C520, U501, R505, J101, R208, R209, R210, R213, Q406, Q402, Q403, C401 , C403, U401 SMT status, MB, J3402, C1714, U1501 SMT status.
2. Measure whether the signals ANGLE_INT_N_3P3, ANGLE_I2C_SDA_3P3, ANGLE_I2C_SCL_3P3, ANGLE_RST_3P3_N are abnormal, and troubleshoot the components on the abnormal signals to replace the problem components.

64M Main Camera Fail

1. Confirm whether the connector (J3302) of Camera RFPC is fully buckled on the MB.
2. Check the connector (J3302) for damage or foreign objects.
3. Swap all keypart confirmation phenomena are related to MB or RFPC.

If the Swap phenomenon is related to MB：

1. Measure whether the impedance of CAM0_DVDD_1P1_MB/ CAM0_AVDD_2P9_MB/ CAM0_AVDD_1P8_MB/CAM02_DRV_AVDD_2P9_MB is abnormal, if it is abnormal, check the component U2210 on the circuit, and replace U2210 if there is a problem.
2. Measure whether the impedance of CAM0_VM_3P1_MB is abnormal, if it is abnormal, check the components C3403, U3402 on the line, if there is a problem, reheat or replace the component (C3403, U3402)
3. Measure CAM012_GYRO_EEPROM_DOVDD_1P8_MB if it is abnormal, check the component U1801 on the line, if there is a problem, replace U1801.

If the swap phenomenon is similar to RFPC

1. Check the connector (CON1602) for damage or foreign objects.
2. Check whether the RFPC itself is damaged (Ex. scratches, screw locks)
3. Turn on the camera module, confirm the phenomenon and the 64M camera module or RFPC

→ If the Swap phenomenon is related to RFPC ：

1. Measure whether the impedance of CAM0_DVDD_1P1_MB/ CAM0_AVDD_2P9_MB/ CAM0_AVDD_1P8_MB/ is abnormal, if abnormal, check the components on the circuit and replace them
2. Check whether the D-phy signal has an open circuit or an abnormal impedance (CAM_MCLK0/CSI0_A0, B0, C0/CSI0_A1, B1C1/CSI0_A2, B2, C2)

12M Wide Camera Fail

1. Check whether the connector (J3302&J3303) of Camera RFPC is fully buckled on the MB.
2. Check whether the connector (J3302&J3303) is damaged or foreign.
3. Swap all keypart confirmation phenomena are related to MB or RFPC.

If Swap phenomenon is related to MB ：

1. Measure whether the impedance of CAM1_VDDD_1P05_MB/ CAM1_VDDA_2P8_MB is abnormal, if it is abnormal, check the component U2210 on the line, if there is a problem, replace U2210.
2. Measure CAM1_VDD_AF_2P8_MB/CAM012_GYRO_EEPROM_DOVDD_1P8_MB, if it is abnormal, check the component U1801 on the line, if there is a problem, replace U1801.

f the swap phenomenon is similar to RFPC:

1. Check the connectors (CON1602, CON1603) for damage or foreign objects.
2. Check whether the RFPC itself is damaged (Ex. scratches, screw locks)
3. Turn on the camera module, confirm the phenomenon and 12M camera module or RFPC

→ If Swap phenomenon is related to RFPC ：

1. Measure whether the impedance of CAM1_VDDD_1P05_MB/ CAM1_VDDA_2P8_MB / CAM1_VDD_AF_2P8/CAM012_GYRO_EEPROM_DOVDD_1P8_MB is abnormal, if it is abnormal, check the components on the circuit and replace them
2. Check whether the MIPI signal impedance is abnormal or abnormal (CAM_MCLK1/CSI1_LANE0_P, M/CSI1_LANE1_P, M/CSI1_LANE2_P, M/CSI1_LANE3_P, M/CSI1_CLK_M, P)

8M Tele Camera Fail

1. Confirm whether the connector (J3301) of Camera RFPC is fully buckled on the MB.
2. Check whether the connector (J3301) is damaged or foreign.
3. Swap all keypart confirmation phenomena are related to MB or RFPC.

If Swap phenomenon is related to MB ：

1. Measure whether the impedance of CAM2_VM_2P8_MB is abnormal. If it is abnormal, check the component U3403 on the line. If there is a problem, replace U3403.
2. Measure CAM2_AVDD_2P8_MB/CAM02_DRV_AVDD_2P9_MB, if it is abnormal, check the component U2210 on the line, if there is a problem, replace U2210.
3. Measure whether the impedance of CAM2_DVDD_1P2_MB_CON is abnormal, if it is abnormal, check the component U2301 on the line, if there is a problem, replace U2301.
4. Measure CAM012_GYRO_EEPROM_DOVDD_1P8_MB if it is abnormal, check the component U1801 on the line, if there is a problem, replace U1801.

If the swap phenomenon is similar to RFPC:

1. Check the connector (CON1601) for damage or foreign matter.
2. Check whether the RFPC itself is damaged (Ex. scratches, screw locks)
3. Turn on the camera module, confirm the phenomenon and 8M camera module or RFPC

→ If Swap phenomenon is related to RFPC ：

1. Measure whether the impedance of CAM2_VM_2P8 / CAM2_DVDD_1P2 / CAM2_DOVDD_1P8/CAM2_AVDD_2P8/CAM2_DRV_AVDD_2P9 is abnormal, if it is abnormal, check the components on the circuit and replace them
2. Check whether the MIPI signal impedance is abnormal or if the impedance is abnormal (CAM_MCLK2/ CSI2_LANE0_M, P/ CSI2_LANE1_M, P/ CSI2_LANE2_M, P/ CSI2_LANE3_M, P/ CSI2_CLK_M, P)

SIM/SD Card Fail (1/3)

1. Make sure that SIMFPC_MID, SIM tray, SIM 1/2, SD card are normal, check the SMT status of SIM SB J01~J05, J106, MB J3401.
2. SIM 1 fail, measure whether SIM SB R108 has power supply (VCC_NFC1), whether R107 has power supply (VREG_L11E_1P8), and whether R110 (UIM1_RESET) is abnormal.
3. SIM 2 fail, measure whether SIM SB R114 has power supply (VCC_NFC2), whether R115 has power supply (VREG_L13E_1P8), and whether R118 (UIM2_RESET) is abnormal.
4. SD card fail, measure whether SIM SB C101 has power supply (VREG_L9C_2P96). If not, check SIM SB J106, D101.
5. Check whether the SIM SB R123 will be pulled from 0V to 1.8V after the SIM tray is inserted, and judge whether the DET pin is active. If not, check the SMT status of SIM SB R123, R122, Q102, R124, R125, J105.
6. Check SDC2_CLK with the SD card inserted, and measure MB C203.1 to see if there is a CLOCK. If not, check the SMT status of MB R201, C203, J3401.
7. Check UIM1_CLK/UIM2_CLK with dual SIM inserted, and measure whether SIM R113 and R120 have CLOCK. If not, check the SMT status of MB U1001, J3401, SIM J106, D114, R120, D110, R113.
8. If there is no problem above, check whether the SIM card used is bad

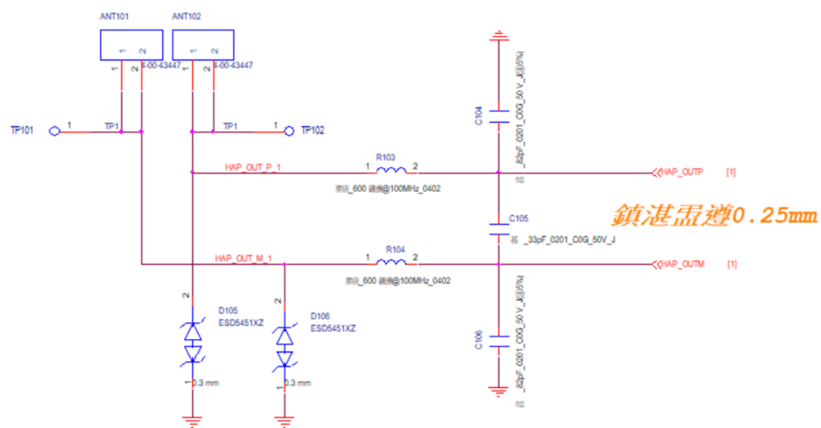
Vibrator Fail

1. Check whether the IO FPC/USB small card connector (J3701, J102) has no buckles or foreign objects or shrapnel is abnormal, and whether there is pressure on the assembly.
2. Measure whether the normal PM impedance of the Vibrator unit is normal without short circuit.
3. Swap mainboard/LKFPC/USB small card to find the problem follow point

If with LKFPC → Replace LKFPC

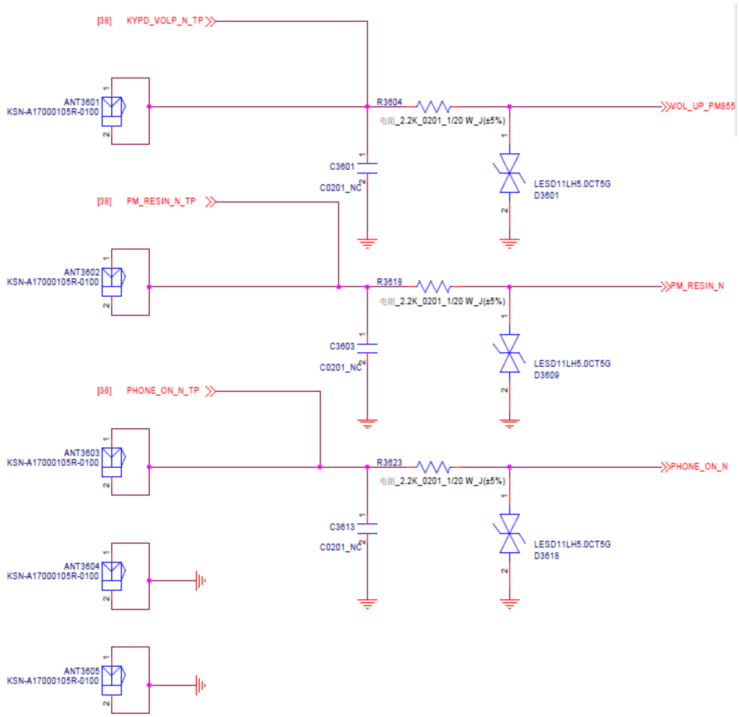
If with USB borad→ Check R103, R104, ANT101, ANT102 and other RLC components SMT status ANT101

If it comes with the mainboard →Check U2601SMT status, first reheat U2601, if it is invalid, replace U2601.



VOL Key Fail

1. Check if ANT3601, ANT3602, ANT3603, ANT3604, ANT3605 and Side key FPC are connected properly.
2. Check the SMT status of MB R3604, R3618, R3623, D3601, D3609, D3618.
3. Press the "DOWM" button to check whether MB C3603.1 has pulled LOW from HIGH.
4. Press the "UP" button to check whether MB C3601.1 pulls LOW from HIGH.



Indicator LED Fail

1. Check the contact status of MB J3701 with USB SB J102, , then confirm problems occurred in USB SB or MB.

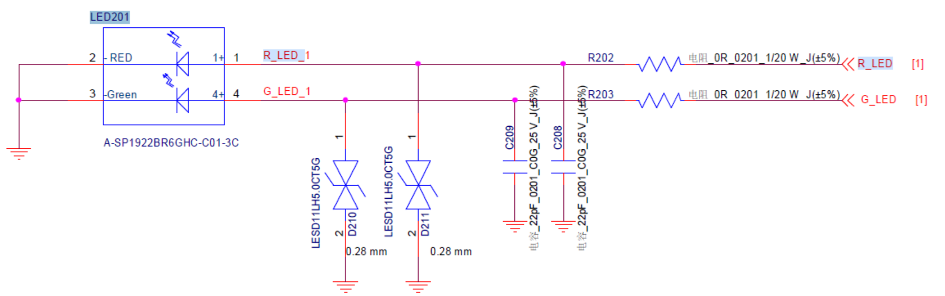
■ If problems occurred in USB SB:

1. Check the SMT status of J102, R202, R203, C208, C209, D210, D211 and LED201.
2. If the above steps are confirmed normal, but the failure is still existing, then swap the LED201.

■ If problems occurred in MB:

1. Check the SMT status of MB J3701 and U1801.
2. If the above steps are confirmed normal, but the failure is still existing, reheat MB U1801 first, then swap the MB U1801.

INDICATOR



Flash LED Fail

1. Check the contact status of MB J3301 with camera FPC CON1601, then confirm problems occurred in Flash FPC, CAM SB or MB.

■ If problems occurred in CAM SB:

1. Check the SMT status of Flash FPC C1601, C1602 and CON1605.
2. Check the voltage of C1601.1 is 3V, if not, check the SMT status of C1601 and C1602.
3. If the above steps are confirmed normal, but the failure is still existing, then swap the Flash FPC.

■ If problems occurred in MB:

1. Check the SMT status of MB J3301, U1801 and C1807.
2. Check the voltage of C1807.1 is 3V, check the SMT status of C1807.
3. If the above steps are confirmed normal, but the failure is still existing, reheat MB U1801 first, then swap the U1801.

Receiver Fail

1. Check the SMT and contact status of Receiver, camera FPC CON1603 and MB J3303.
2. Check the SMT and contact status of CAM SB AU7301 and AU7302 between Receiver.
3. Check the SMT status of MB R3003, R3004, R3005, R3006, L3002, L3003, U3001.
4. Check the voltage of MB C3012.2 is 1.8V. If is not, check the SMT status.
5. Check the voltage of MB C3001.1 is normal. If not, check the SMT status of L3001.
6. If the above steps are confirmed normal, but the failure is still existing. Reheat MB U3001 first, then swap the MB U3001.

Speaker Fail

1. Check the SMT and contact status of Speaker, IO FPC, SIM FPC1 and SIM FPC2.
2. Check the USB SB ANT201 and ANT202 contact status with Speaker.
3. Check the SMT status of USB SB: B201, B202, J102，SIM SB: R202, R203, R204, R205, B201, B202, U202. MB: J3701, ANT3001, ANT3002.
4. Check the voltage of SIM SB C211.2 is 1.8V. If is not, check the SMT and contact status of FPCs between SIM board and MB.
5. Check the voltage of SIM SB C201.1 is normal. If is not, check the SMT status of L201. If the above steps are confirmed normal, but the failure is still existing. Reheat SIM SB U202 first, then swap the SIM SB U202.

Main Mic Fail

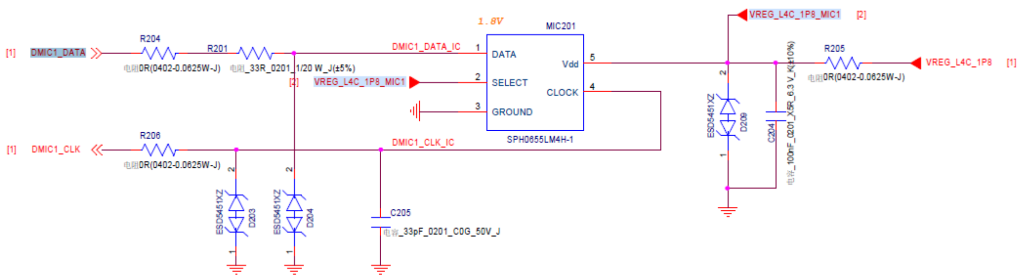
■ If problems occurred in MB:

1. Check the SMT status of MB J3701, R326.
2. Check the voltage of C2008.1 is 1.8V, if is not or abnormal, reheat U1801 first, then swap the U1801.

■ If problems occurred in USB SB:

1. Check the SMT status of USB SB J102
2. Check the SMT status of USB SB MIC201, R204, R205, R206, R201, D203, D204, C205, C204, D209.
3. Check the voltages of USB SB R205.1 / R205.2 are 1.8V.
4. If the above steps are confirmed normal, but the failure is still existing.
5. Reheat the USB SB MIC201 first, then swap the MB MIC201.

Main DMIC



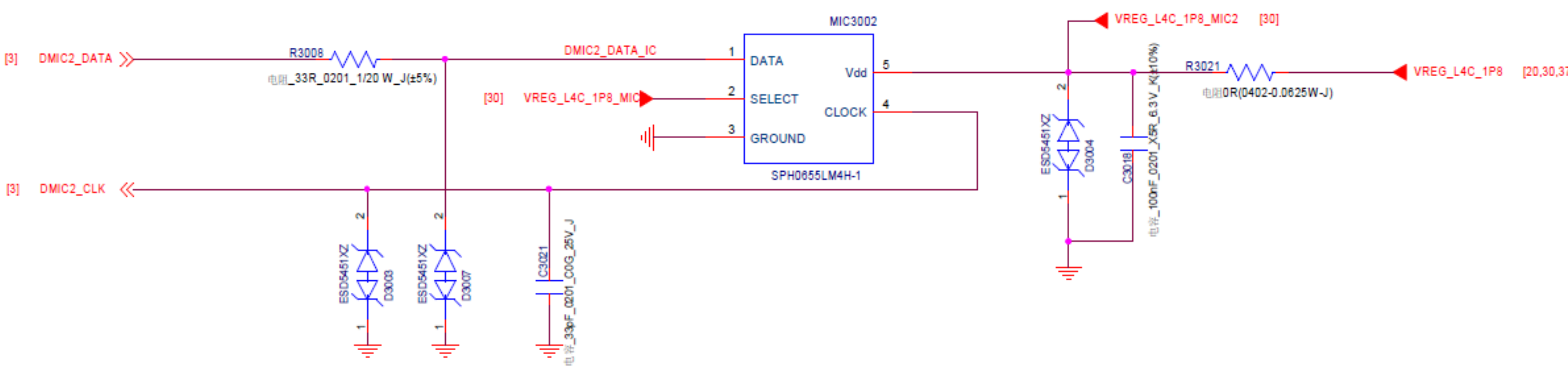
Slave Mic Fail

1. Check the voltage of MB C2008.1 is 1.8V, if is not or abnormal, reheat U1801, the failure is still existing, please swap the U1801.
2. Check the SMT status of MB MIC3002, R327, R3008, D3003, D3004, D3007, C3018, C3021, R3021.
3. Check the voltages of MB R3021.1 / R3021.2 are 1.8V.

If the above steps are confirmed normal, but the failure is still existing.

reheat MB MIC3002 first, then swap the MB MIC3002.

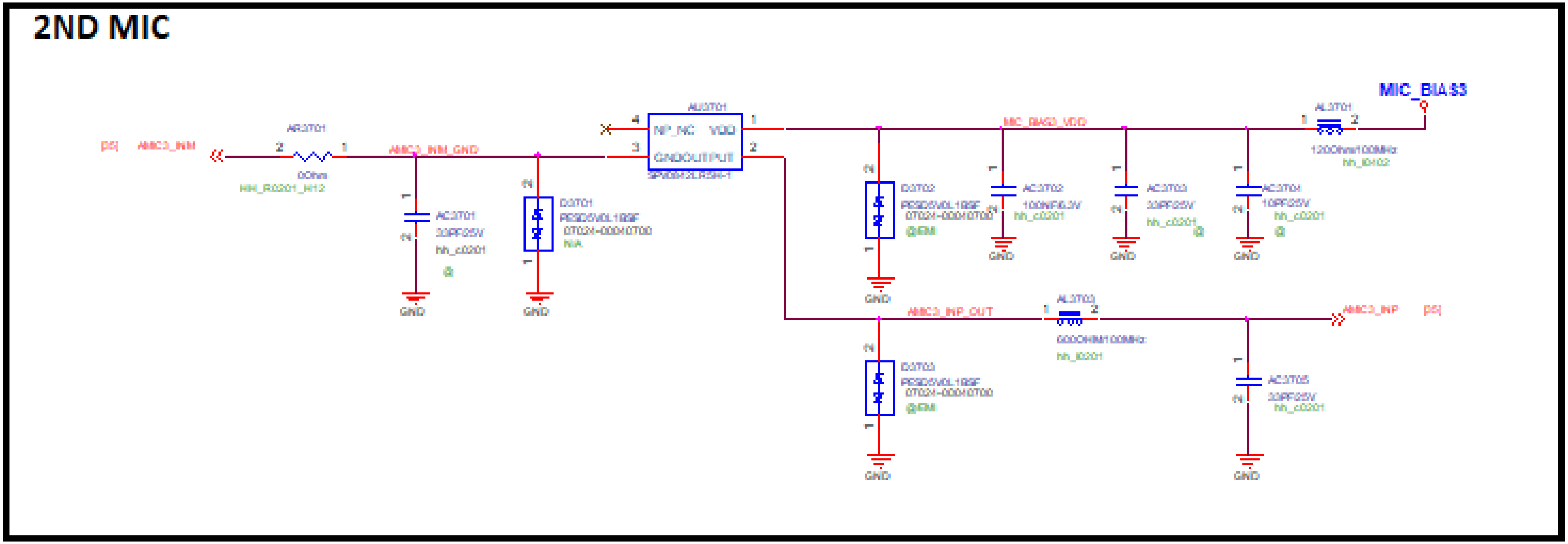
Slave DMIC



3rd Mic Fail

1. Check the voltage of MB C2008.1 whether there is power supply, if is not or abnormal, reheat U1801, the failure is still existing, please swap the U1801.
2. Check the SMT status of MB MIC3001, R328, R3014, D3005, D3006, D3008, C3029, C3032, R3017.
3. Check the voltages of MB R3017.1/ R3017.2 are 1.8V。
4. If the above steps are confirmed normal, but the failure is still existing.

   reheat MB MIC3001 first, and then swap MB MIC3001。



Headset Fail

1. Check the status of IO FPC and connectors, then confirm problems occurred in USB SB or MB.

■ If problems occurred in USB SB:

1. Check the SMT status of USB SB J101, J102, R101, R102, D103, D104, Z101, C101, C102, C103.
2. If the above steps are confirmed normal, but the failure is still existing. Please swap USB SB Z101.

■ If problems occurred in MB:

1. Check the SMT status of MB J3701, R3713, R3714, D3703, D3704, U3702, R3708, R3710, C3710.
2. Check the voltage of MB C3710.2 is 3.1V。
3. If the above steps are confirmed normal, but the failure is still existing. Please reheat MB U3702 and U201, then swap MB U3702, U201。